

박수민_0629_ds1302_burst코드

#main.c

/*

* 10.MY_DS1302_ver2.c

*

* Created: 2026-06-29 오후 4:55:35

* Author : kccistc

*/

#define F_CPU 16000000UL

#include <avr/io.h>

#include <avr/interrupt.h> // sei, ISR 등

#include <stdio.h>

#include <util/delay.h>

extern void init_uart0();

extern void UART0_transmit(*uint8_t* data);

// extern void dht11_main();

extern void ds1302_main();

FILE OUTPUT = *FDEV_SETUP_STREAM*(UART0_transmit, *NULL*, *_FDEV_SETUP_WRITE*);

int main(void)

{

init_uart0();

stdout = &OUTPUT; // printf가 동작할 수 있도록 stdout을 설정

ds1302_main();

while (1)

{

//dht11_main();

//_delay_ms(1500); // DHT11 안정화 시간

}

}

```

#ds1302.c
/*
 * ds1301.c
 *
 * Created: 2026-06-26 오후 2:42:13
 * Author: kccistc
 */

#include "ds1302.h"
#include <avr/interrupt.h>

void init_ds1302(t_ds1302 *p);
void init_date_time(t_ds1302 *p);
void init_ddr_ds1302(void);
void init_gpio_ds1302(void);

void ds1302_main(void);
void clock_ds1302(void);
void read_time_ds1302(t_ds1302 *p);
void read_date_ds1302(t_ds1302 *p);
void tx_ds1302(uint8_t data);
void rx_1302(uint8_t *pdata8bits);
void write_ds1302(uint8_t addr, uint8_t data);
void rx_burst_1302(uint8_t *pdata8bits);
void read_burst_ds1302(t_ds1302 *p);

uint8_t read_ds1302(uint8_t addr);
uint8_t dec2bcd(uint8_t data);
uint8_t bcd2dec(uint8_t data);

extern void pc_command_processing(t_ds1302 *p);

void ds1302_main(void)
{
    t_ds1302 rtc;

    init_date_time(&rtc);
    init_ddr_ds1302();
    init_gpio_ds1302();           // all LOW로 설정
    init_ds1302(&rtc);
    sei();    // UART 인터럽트 활성화
}

```

```

while (1)
{
    pc_command_processing(&rtc);

    read_burst_ds1302(&rtc);

    ///// 1. read time
    //read_time_ds1302(&rtc);
    //
    ///// 2. read date
    //read_date_ds1302(&rtc);

    // 3. printf date & time
    printf("DATE: %d-%d-%d ", rtc.year, rtc.month, rtc.date);
    printf("TIME: %d:%d:%d\n", rtc.hour, rtc.minutes, rtc.seconds);

    // 4. delay_ms(1000);
    _delay_ms(1000);
}

};

void read_time_ds1302(t_ds1302 *p)
{
    p->seconds = read_ds1302(ADDR_SECONDS);
    p->minutes = read_ds1302(ADDR_MINUTES);
    p->hour     = read_ds1302(ADDR_HOUR);
}

void read_date_ds1302(t_ds1302 *p)
{
    p->date  = read_ds1302(ADDR_DATE);
    p->month = read_ds1302(ADDR_MONTH);
    p->year  = read_ds1302(ADDR_YEAR);
}

uint8_t read_ds1302(uint8_t addr)
{
    uint8_t data8bits = 0;    // 1bit씩 읽어서 답을 변수

    // 4. CE low
    // 5. return(bcd to dec)

```

```

// 1. CE low --> High
DS1302_RST_PORT |= 1 << DS1302_RST;

// 2. ADDR 전송
tx_ds1302(addr+1);      // read addr

// 3. DATA 읽어들이기
rx_1302(&data8bits);

// 4. CE HIGH ---> LOW
DS1302_RST_PORT &= ~(1 << DS1302_RST);

return bcd2dec(data8bits);
}

void rx_1302(uint8_t *pdata8bits)
{
    uint8_t temp=0;

    // 1. 입력 모드로 설정
    DS1302_DAT_DDR &= ~(1 << DS1302_DAT);           // write mode
    _delay_us(2);

    // LSB로부터 차례로
    for (int i=0; i<8; i++)
    {
        if (DS1302_DAT_PIN & (1 << DS1302_DAT))
        {
            temp |= 1 << i;           // 1의 조건만 set
        }

        // 3. CLK을 high -> low
        if (i != 7)                   // 마지막 비트를 읽을 때는 clk
            clock_ds1302();           을 보내주지 않는다.

        *pdata8bits = temp;
    }
};

void init_ds1302(t_ds1302 *p)
{

```

```

        write_ds1302(ADDR_WRITEPROTECTED, 0x00);

        write_ds1302(ADDR_SECONDS,   p->seconds);
        write_ds1302(ADDR_MINUTES,   p->minutes);
        write_ds1302(ADDR_HOUR,       p->hour);
        write_ds1302(ADDR_DATE,        p->date);
        write_ds1302(ADDR_MONTH,       p->month);
        write_ds1302(ADDR_DAYOFWEEK,   p->dayofweek);
        write_ds1302(ADDR_YEAR,        p->year);
    }

```

```

// decimal --> bcd
// ex) 25
//      dec                bcd
//      00011001          0010 0101

```

```

uint8_t dec2bcd(uint8_t data)
{
    uint8_t high, low;

    high = (data/10) << 4;  // high nibble에 위치
    low = data % 10;

    return (high+low);
}

```

```

// bcd --> decimal
// 1. 입력 bcd
// ex. 26년의 bcd
// 7654 3210
// 0010 0110
//      2          6
// x10   x1
// =====
// 26

```

```

uint8_t bcd2dec(uint8_t data)
{
    uint8_t high, low;

    low = data & 0x0f;

```

```

        high = (data >> 4) * 10;

        return (high + low);
    }

void write_ds1302(uint8_t addr, uint8_t data)
{
    // 1. CE low --> High
    DS1302_RST_PORT |= 1 << DS1302_RST;

    // 2. ADDR 전송
    tx_ds1302(addr);

    // 3. DATA 전송
    tx_ds1302(dec2bcd(data));

    // 4. CE HIGH ----> LOW
    DS1302_RST_PORT &= ~(1 << DS1302_RST);
}

void tx_ds1302(uint8_t data)
{
    // 1. 출력 모드로 설정
    DS1302_DAT_DDR |= 1 << DS1302_DAT;           // write mode
    // ex) 0x80
    //           M           L
    //           1000 000
    for (int i=0; i<8; i++)
    {
        // 2. 1 bit를 전송
        if (data & (1 << i))
            DS1302_DAT_PORT |= 1 << DS1302_DAT;           // 1
        else DS1302_DAT_PORT &= ~(1 << DS1302_DAT);       // 0

        // 3. CLK을 high -> low
        clock_ds1302();
    }
}

void clock_ds1302(void)
{
    // LOW ----> HIGH ----> LOW

```

```

    DS1302_CLK_PORT &= ~(1<<DS1302_CLK);
    _delay_us(1);
    DS1302_CLK_PORT |= 1<<DS1302_CLK;
    _delay_us(1);
    DS1302_CLK_PORT &= ~(1<<DS1302_CLK);
    _delay_us(1);
}

void init_ddr_ds1302(void)
{
    DDRF &= ~(1 << DS1302_CLK | 1 << DS1302_DAT | 1 << DS1302_RST);
    DDRF |= 1 << DS1302_CLK | 1 << DS1302_DAT | 1 << DS1302_RST;    // 출력
    력 모드로 설정
}

void init_gpio_ds1302(void)
{
    DS1302_CLK_PORT &= ~( 1 << DS1302_CLK | 1 << DS1302_DAT | 1 <<
DS1302_RST);
    _delay_ms(2);
}

void init_date_time(t_ds1302 *p)
{
    p->year      = 26;
    p->month      = 06;
    p->date       = 29;
    p->dayofweek  = 6;
    p->hour       = 10;
    p->minutes    = 02;
    p->seconds    = 00;
}

void rx_burst_1302(uint8_t *pdata8bits)
{
    uint8_t temp = 0;

    DS1302_DAT_DDR &= ~(1 << DS1302_DAT);
    _delay_us(2);

```

```

    for (int i = 0; i < 8; i++)
    {
        if (DS1302_DAT_PIN & (1 << DS1302_DAT))
            temp |= 1 << i;

        clock_ds1302(); // ← 마지막 비트도 CLK 보냄 (다음 바이트 준비)
    }
    *pdata8bits = temp;
}

void read_burst_ds1302(t_ds1302 *p)
{
    uint8_t data[7];

    DS1302_RST_PORT |= 1 << DS1302_RST;
    _delay_us(5);

    tx_ds1302(0xBF);

    rx_burst_1302(&data[0]);
    rx_burst_1302(&data[1]);
    rx_burst_1302(&data[2]);
    rx_burst_1302(&data[3]);
    rx_burst_1302(&data[4]);
    rx_burst_1302(&data[5]);
    rx_1302(&data[6]); // 마지막은 기존 함수

    DS1302_RST_PORT &= ~(1 << DS1302_RST);

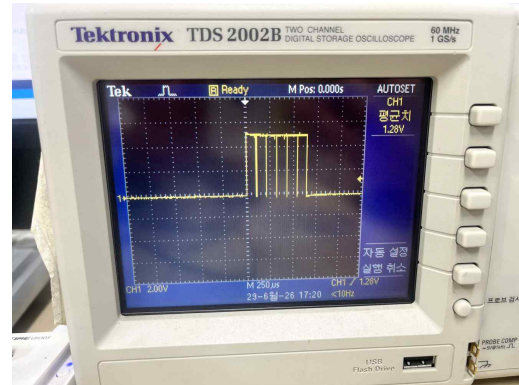
    p->seconds = bcd2dec(data[0] & 0x7F);
    p->minutes = bcd2dec(data[1] & 0x7F);
    p->hour = bcd2dec(data[2] & 0x3F);
    p->date = bcd2dec(data[3] & 0x3F);
    p->month = bcd2dec(data[4] & 0x1F);
    p->dayofweek = bcd2dec(data[5] & 0x07);
    p->year = bcd2dec(data[6]);
}

```


오실로스코프 비교

기존 방식(개별읽기)

CE ↑ ↓ → seconds (트랜잭션 1)
CE ↑ ↓ → minutes (트랜잭션 2)
CE ↑ ↓ → hour (트랜잭션 3)
CE ↑ ↓ → date (트랜잭션 4)
CE ↑ ↓ → month (트랜잭션 5)
CE ↑ ↓ → year (트랜잭션 6)
→ CE 토글 6번 => 펄스 6개 보임



BURST

CE ↑ → 0xBF 명령 전송
→ seconds 수신
→ minutes 수신
→ hour 수신
→ date 수신
→ month 수신
→ year 수신

CE ↓

→ CE 토글 1번 => 펄스 1개 보임

